W2026 MAD317-01 Mobile App Project Development (Kayode Olukayode)

Final Project Report and Reflection

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Repository: https://github.com/darshil2711/Study-TODO

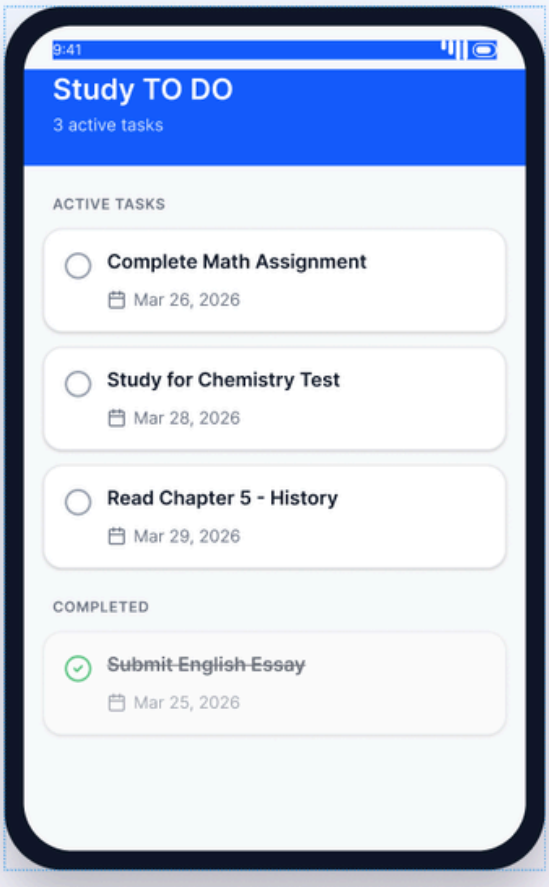
1. Project Overview

- Application Name:
  - Study TO DO
- Problem Statement:
  - Students often struggle to manage multiple assignments, deadlines, and study tasks due to lack of organization and reliance on scattered tools. This results in missed deadlines and increased stress.
  - The Study TO DO application solves this problem by providing a centralized mobile platform where users can create, manage, and track their study tasks efficiently.
- Target Users:
  - The primary users are college students who need a simple and effective way to manage academic tasks. These users typically rely on smartphones and prefer quick and easy task management solutions.
- Platform:
  - The application was developed using Flutter (cross-platform) targeting Android devices.

2. Design Process (Assignment 3 Summary)

The design process focused on creating a simple, user-friendly interface following Android Material Design guidelines.

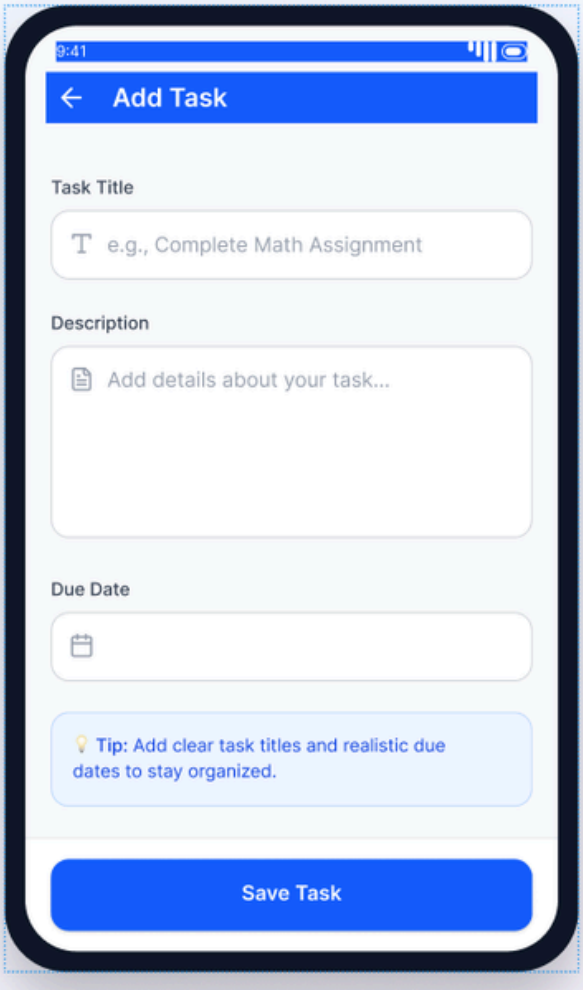
- UI/UX Approach:
  - The application uses a minimal and clean design to reduce cognitive load. Navigation is simple and intuitive, allowing users to quickly access key features such as adding and viewing tasks.
- Key Design Decisions:
  - Use of Floating Action Button (+) for adding tasks
  - Card-based layout for displaying tasks
  - Clear typography hierarchy for readability
  - Minimal color scheme to maintain focus
- Wireframes & Prototype:
  - Low-fidelity wireframes were created to define layout structure, followed by high-fidelity mockups in Figma. An interactive prototype was developed to validate navigation flow.



Controls

Clear All Tasks (Show Empty State)

Reset Sample Tasks



Material Design Features

- ✅ App Bar with back navigation
- ✅ Input fields with icons
- ✅ Focus states with blue ring
- ✅ Proper touch targets (48dp+)
- ✅ 16dp spacing between elements
- ✅ Rounded corners (12dp)
- ✅ Fixed action button at bottom
- ✅ Clean, readable typography

Form validates required fields before saving

3. MVP Development

The Study TO DO application was developed as a functional MVP using a structured architecture.

- Architecture Used:
  - The application follows the MVVM (Model-View-ViewModel) architecture, ensuring separation of concerns between UI, business logic, and data handling.
- Core Features Implemented:
  - Add new study tasks
  - View list of tasks
  - View task details
  - Delete tasks
  - Mark tasks as completed
- Technology Stack:
  - Flutter (UI framework)
  - Dart (programming language)
  - Android Emulator for testing

4. Testing & Debugging

4.1 Testing Strategy

The application was tested using multiple approaches:

- Functional testing
- Usability testing
- Performance testing
- Error handling validation

Test ID	Scenario	Expected Result	Status
TC01	Open App	Home screen loads	Pass
TC02	Add Task	Task appears in list	Pass
TC03	Empty Input	Error message shown	Pass
TC04	Delete Task	Task removed	Pass

Bug ID	Description	Severity	Status
BUG01	Empty task allowed	Medium	Pass
BUG02	List not refreshing	Medium	Pass
BUG03	No confirmation message	Low	Pass

4.4 Debugging Approach
Debugging was performed using console logs and runtime testing in the emulator. Issues were identified by tracking application behavior and verifying user inputs.

- Fixes included:
  - Adding input validation
  - Updating UI refresh logic
  - Improving task handling functions

5. Final Build & Deployment Readiness
5.1 Code Refactoring
The codebase was cleaned and reorganized to improve readability and maintainability. Unused files and redundant code were removed.

5.2 Performance Optimization
Performance improvements were achieved by:

- Using efficient UI components (ListView.builder)
- Reducing unnecessary widget rebuilds

This resulted in smoother UI performance and faster screen transitions.

5.3 Security Implementation
Security measures included:

- Input validation to prevent invalid data
- Avoiding storage of sensitive information
- Basic error handling to prevent crashes

5.4 Build & Deployment
The application was prepared for deployment by generating a release APK using Flutter.

- flutter build apk --release

The application was successfully installed and tested on an Android device.
Deployment Readiness Checklist:

- App name: Study TO DO
- Description: Study task management app
- Screenshots included
- No sensitive permissions required

6. Challenges & Solutions

Challenge	Solution	Outcome
UI alignment issues	Adjusted layout spacing	Improved design consistency
Task not saving properly	Fixed logic in controller	Feature works correctly
Navigation errors	Implemented proper routing	Smooth navigation

7. Lessons Learned
This project helped develop practical skills in mobile application development.

- Technical Skills:
  - Flutter development
  - UI design implementation
  - Debugging and testing
- Development Insights:
  - Importance of structured planning
  - Need for proper testing before final build
  - Value of clean code practices

8. Future Improvements

The application can be enhanced with:

- Task reminders and notifications
- Cloud synchronization
- User authentication
- Dark mode support

9. Personal Reflection

During this project, I worked as the sole developer responsible for design, development, testing, and deployment.

This project improved both technical and soft skills, including:

- Problem-solving
- Time management
- Debugging techniques

It also provided real-world experience in developing a mobile application from concept to deployment.